Kunwar Ashwani Pratap Singh

BCADS-14

1250258250

CLASS ROLL NO : 11

SUBJECT : DATA VISUALIZATION

```
import pandas as pd
import matplotlib.pyplot as plt

matches = pd.read_csv(r"matches.csv")
deliveries = pd.read_csv(r"deliveries.csv")

matches

        id  season           city        date
team1  \
0        1    2017      Hyderabad  2017-04-05          Sunrisers
Hyderabad
1        2    2017           Pune  2017-04-06               Mumbai
Indians
2        3    2017         Rajkot  2017-04-07                Gujarat
Lions
3        4    2017         Indore  2017-04-08       Rising Pune
Supergiant
4        5    2017      Bangalore  2017-04-08  Royal Challengers
Bangalore
..     ...     ...            ...         ...
...
751  11347    2019         Mumbai    05/05/19        Kolkata Knight
Riders
752  11412    2019        Chennai    07/05/19          Chennai Super
Kings
753  11413    2019  Visakhapatnam    08/05/19          Sunrisers
Hyderabad
754  11414    2019  Visakhapatnam    10/05/19               Delhi
Capitals
755  11415    2019      Hyderabad    12/05/19               Mumbai
Indians

                           team2                   toss_winner
toss_decision  \
0    Royal Challengers Bangalore  Royal Challengers Bangalore
field
1         Rising Pune Supergiant       Rising Pune Supergiant
```

```
field
2          Kolkata Knight Riders        Kolkata Knight Riders
field
3                Kings XI Punjab              Kings XI Punjab
field
4               Delhi Daredevils  Royal Challengers Bangalore
bat
..                           ...                          ...
...
751               Mumbai Indians               Mumbai Indians
field
752               Mumbai Indians          Chennai Super Kings
bat
753               Delhi Capitals               Delhi Capitals
field
754          Chennai Super Kings          Chennai Super Kings
field
755          Chennai Super Kings               Mumbai Indians
bat

     result  dl_applied                       winner  win_by_runs  \
0    normal           0          Sunrisers Hyderabad           35
1    normal           0       Rising Pune Supergiant            0
2    normal           0        Kolkata Knight Riders            0
3    normal           0              Kings XI Punjab            0
4    normal           0  Royal Challengers Bangalore           15
..      ...         ...                          ...          ...
751  normal           0               Mumbai Indians            0
752  normal           0               Mumbai Indians            0
753  normal           0               Delhi Capitals            0
754  normal           0          Chennai Super Kings            0
755  normal           0               Mumbai Indians            1

     win_by_wickets player_of_match  \
0                 0    Yuvraj Singh
1                 7       SPD Smith
2                10         CA Lynn
3                 6      GJ Maxwell
4                 0       KM Jadhav
..              ...             ...
751               9       HH Pandya
752               6        AS Yadav
753               2         RR Pant
754               6    F du Plessis
755               0       JJ Bumrah

                                          venue         umpire1  \
0    Rajiv Gandhi International Stadium, Uppal     AY Dandekar
1      Maharashtra Cricket Association Stadium  A Nand Kishore
2       Saurashtra Cricket Association Stadium     Nitin Menon
```

```
3                       Holkar Cricket Stadium    AK Chaudhary
4                        M Chinnaswamy Stadium             NaN
..                                         ...             ...
751                           Wankhede Stadium   Nanda Kishore
752                  M. A. Chidambaram Stadium     Nigel Llong
753                           ACA-VDCA Stadium             NaN
754                           ACA-VDCA Stadium   Sundaram Ravi
755         Rajiv Gandhi Intl. Cricket Stadium     Nitin Menon

            umpire2                   umpire3
0          NJ Llong                       NaN
1            S Ravi                       NaN
2         CK Nandan                       NaN
3     C Shamshuddin                       NaN
4               NaN                       NaN
..              ...                       ...
751        O Nandan                   S Ravi
752     Nitin Menon                 Ian Gould
753             NaN                       NaN
754  Bruce Oxenford  Chettithody Shamshuddin
755       Ian Gould               Nigel Llong

[756 rows x 18 columns]

deliveries

        match_id  inning         batting_team
bowling_team  \
0              1       1  Sunrisers Hyderabad  Royal Challengers
Bangalore
1              1       1  Sunrisers Hyderabad  Royal Challengers
Bangalore
2              1       1  Sunrisers Hyderabad  Royal Challengers
Bangalore
3              1       1  Sunrisers Hyderabad  Royal Challengers
Bangalore
4              1       1  Sunrisers Hyderabad  Royal Challengers
Bangalore
...          ...     ...                  ...
...
179073     11415       2  Chennai Super Kings               Mumbai
Indians
179074     11415       2  Chennai Super Kings               Mumbai
Indians
179075     11415       2  Chennai Super Kings               Mumbai
Indians
179076     11415       2  Chennai Super Kings               Mumbai
Indians
179077     11415       2  Chennai Super Kings               Mumbai
Indians
```

```
        over  ball    batsman non_striker      bowler
is_super_over  ...  \
0          1     1  DA Warner    S Dhawan    TS Mills
0  ...
1          1     2  DA Warner    S Dhawan    TS Mills
0  ...
2          1     3  DA Warner    S Dhawan    TS Mills
0  ...
3          1     4  DA Warner    S Dhawan    TS Mills
0  ...
4          1     5  DA Warner    S Dhawan    TS Mills
0  ...
...      ...   ...        ...         ...         ...             ...
...
179073    20     2  RA Jadeja   SR Watson  SL Malinga
0  ...
179074    20     3  SR Watson   RA Jadeja  SL Malinga
0  ...
179075    20     4  SR Watson   RA Jadeja  SL Malinga
0  ...
179076    20     5  SN Thakur   RA Jadeja  SL Malinga
0  ...
179077    20     6  SN Thakur   RA Jadeja  SL Malinga
0  ...

        bye_runs  legbye_runs  noball_runs  penalty_runs  batsman_runs
\
0              0            0            0             0             0

1              0            0            0             0             0

2              0            0            0             0             4

3              0            0            0             0             0

4              0            0            0             0             0

...          ...          ...          ...           ...           ...

179073         0            0            0             0             1

179074         0            0            0             0             2

179075         0            0            0             0             1

179076         0            0            0             0             2

179077         0            0            0             0             0
```

```
        extra_runs  total_runs  player_dismissed dismissal_kind
fielder
0                0           0               NaN            NaN
NaN
1                0           0               NaN            NaN
NaN
2                0           4               NaN            NaN
NaN
3                0           0               NaN            NaN
NaN
4                2           2               NaN            NaN
NaN
...            ...         ...               ...            ...
...
179073           0           1               NaN            NaN
NaN
179074           0           2               NaN            NaN
NaN
179075           0           1         SR Watson        run out  KH
Pandya
179076           0           2               NaN            NaN
NaN
179077           0           0         SN Thakur            lbw
NaN

[179078 rows x 21 columns]
```

```
# i) Match won by the maximum margin of runs
matches.loc[matches['win_by_runs'].idxmax(),
            ['season','team1','team2','winner','win_by_runs']]
```

```
season                      2017
team1            Mumbai Indians
team2          Delhi Daredevils
winner           Mumbai Indians
win_by_runs                 146
Name: 43, dtype: object
```

```
# ii) Match won by maximum wickets
matches.loc[matches['win_by_wickets'].idxmax(),
            ['season','team1','team2','winner','win_by_wickets']]
```

```
season                              2017
team1                     Gujarat Lions
team2             Kolkata Knight Riders
winner            Kolkata Knight Riders
win_by_wickets                       10
Name: 2, dtype: object
```

```python
# iii) Match won by the minimum margin of runs
matches[matches['win_by_runs'] > 0].loc[
    matches[matches['win_by_runs'] > 0]['win_by_runs'].idxmin(),
    ['season','team1','team2','winner','win_by_runs']
]
```

```
season                              2017
team1                    Mumbai Indians
team2            Rising Pune Supergiant
winner                   Mumbai Indians
win_by_runs                           1
Name: 58, dtype: object
```

```python
# iv) Match won by minimum wickets
matches[matches['win_by_wickets'] > 0].loc[
    matches[matches['win_by_wickets'] > 0]['win_by_wickets'].idxmin(),
    ['season','team1','team2','winner','win_by_wickets']
]
```

```
season                                2015
team1                    Kings XI Punjab
team2              Kolkata Knight Riders
winner             Kolkata Knight Riders
win_by_wickets                         1
Name: 559, dtype: object
```

```python
# v) Matches where D/L method was and wasn't applied
matches['dl_applied'].value_counts()
```

```
dl_applied
0    737
1     19
Name: count, dtype: int64
```

```python
# vi) No. of matches held in each city
matches['city'].value_counts()
```

```
city
Mumbai            101
Kolkata            77
Delhi              74
Bangalore          66
Hyderabad          64
Chennai            57
Jaipur             47
Chandigarh         46
Pune               38
Durban             15
Bengaluru          14
Visakhapatnam      13
Centurion          12
```

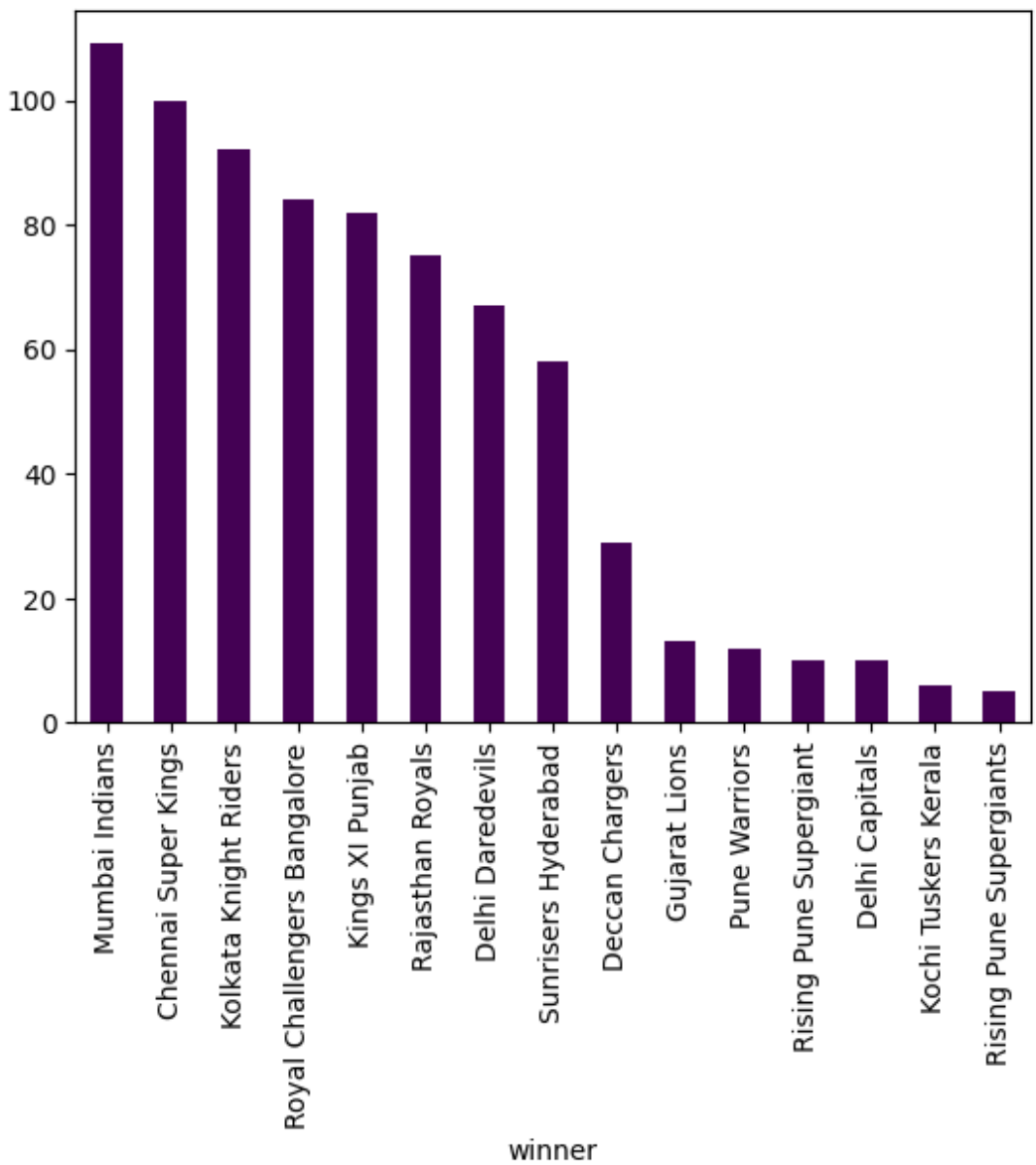
```
Ahmedabad           12
Mohali              10
Rajkot              10
Dharamsala           9
Indore               9
Johannesburg         8
Port Elizabeth       7
Abu Dhabi            7
Cape Town            7
Ranchi               7
Cuttack              7
Sharjah              6
Raipur               6
Kochi                5
Kanpur               4
Kimberley            3
Nagpur               3
East London          3
Bloemfontein         2
Name: count, dtype: int64
```

```
# vii) No. of matches won by each team
matches['winner'].value_counts()
```

```
winner
Mumbai Indians                  109
Chennai Super Kings             100
Kolkata Knight Riders            92
Royal Challengers Bangalore      84
Kings XI Punjab                  82
Rajasthan Royals                 75
Delhi Daredevils                 67
Sunrisers Hyderabad              58
Deccan Chargers                  29
Gujarat Lions                    13
Pune Warriors                    12
Rising Pune Supergiant           10
Delhi Capitals                   10
Kochi Tuskers Kerala              6
Rising Pune Supergiants           5
Name: count, dtype: int64
```

```
matches['winner'].value_counts().plot(kind='bar', colormap='viridis')
```

```
<Axes: xlabel='winner'>
```




```
# viii) No. of matches held every season
matches['season'].value_counts().sort_index()
```

```
season
2008    58
2009    57
2010    60
2011    73
2012    74
2013    76
2014    60
2015    59
2016    60
2017    59
```

```
2018    60
2019    60
Name: count, dtype: int64
```

```
# ix) Top 10 players based on no. of Man of Match (MOM) awards won
matches['player_of_match'].value_counts().head(10)
```

```
player_of_match
CH Gayle          21
AB de Villiers    20
MS Dhoni          17
RG Sharma         17
DA Warner         17
YK Pathan         16
SR Watson         15
SK Raina          14
G Gambhir         13
MEK Hussey        12
Name: count, dtype: int64
```

```
# x) Does winning the toss mean winning the match?
toss_win = matches[matches['toss_winner'] ==
matches['winner']].shape[0]
total_matches = matches.shape[0]
(toss_win / total_matches) * 100
```

```
51.98412698412699
```

```
# xi) What was the decision taken by captains when they won the toss?
matches['toss_decision'].value_counts()
```

```
toss_decision
field    463
bat      293
Name: count, dtype: int64
```

```
# xii) No. of matches where D/L method was applied every season
matches[matches['dl_applied'] == 1]
['season'].value_counts().sort_index()
```

```
season
2008    2
2009    3
2011    3
2014    1
2015    2
2016    4
2017    1
2018    3
Name: count, dtype: int64
```

```python
# xiii) Different results for matches
matches['result'].value_counts()
```

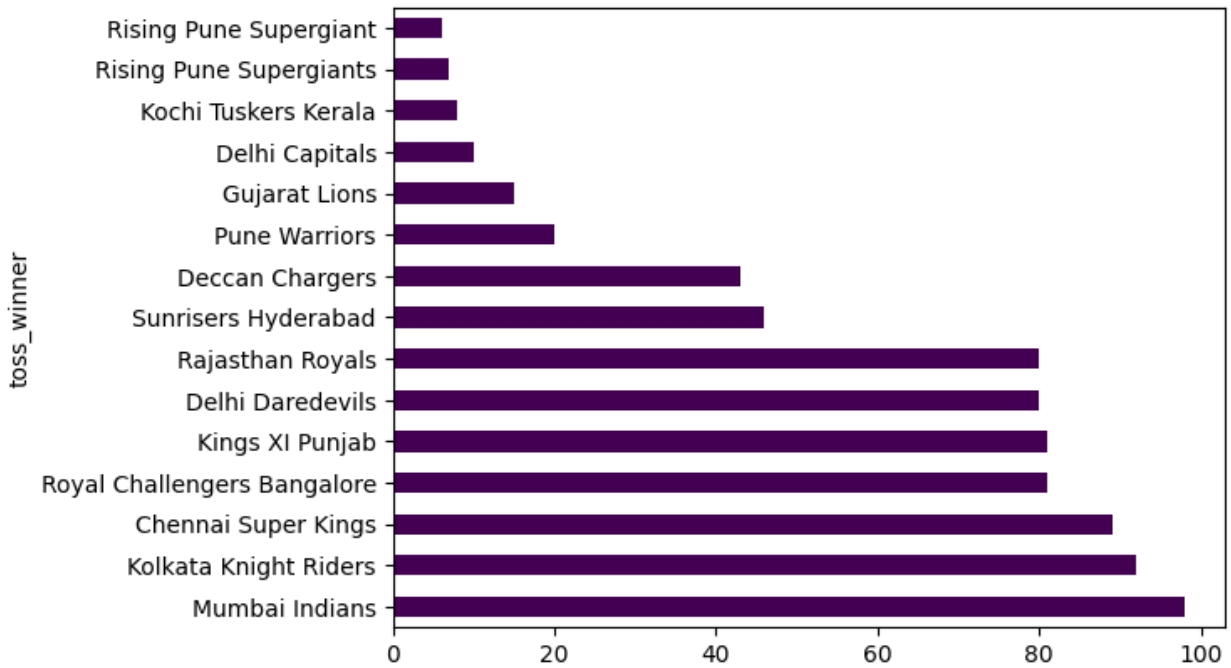
```
result
normal       743
tie            9
no result      4
Name: count, dtype: int64
```

```python
# xiv) How many times did a team win the toss?
matches['toss_winner'].value_counts()
```

```
toss_winner
Mumbai Indians                  98
Kolkata Knight Riders           92
Chennai Super Kings             89
Kings XI Punjab                 81
Royal Challengers Bangalore     81
Delhi Daredevils                80
Rajasthan Royals                80
Sunrisers Hyderabad             46
Deccan Chargers                 43
Pune Warriors                   20
Gujarat Lions                   15
Delhi Capitals                  10
Kochi Tuskers Kerala             8
Rising Pune Supergiants          7
Rising Pune Supergiant           6
Name: count, dtype: int64
```

```python
matches['toss_winner'].value_counts().plot(kind='barh',
colormap='viridis')
```

```
<Axes: ylabel='toss_winner'>
```




```
# xv) Best venue for defending a total
matches[matches['win_by_runs'] > 0]['venue'].value_counts().head(1)

# Best venue for chasing a total
matches[matches['win_by_wickets'] > 0]['venue'].value_counts().head(1)
```

```
venue
Eden Gardens    45
Name: count, dtype: int64
```

```
# xvi) Best defending team
matches[matches['win_by_runs'] > 0]['winner'].value_counts().head(1)

# Best chasing team
matches[matches['win_by_wickets'] > 0]
['winner'].value_counts().head(1)
```

```
winner
Kolkata Knight Riders    56
Name: count, dtype: int64
```

```
# xvii) No. of matches played in different stadiums
matches['venue'].value_counts()
```

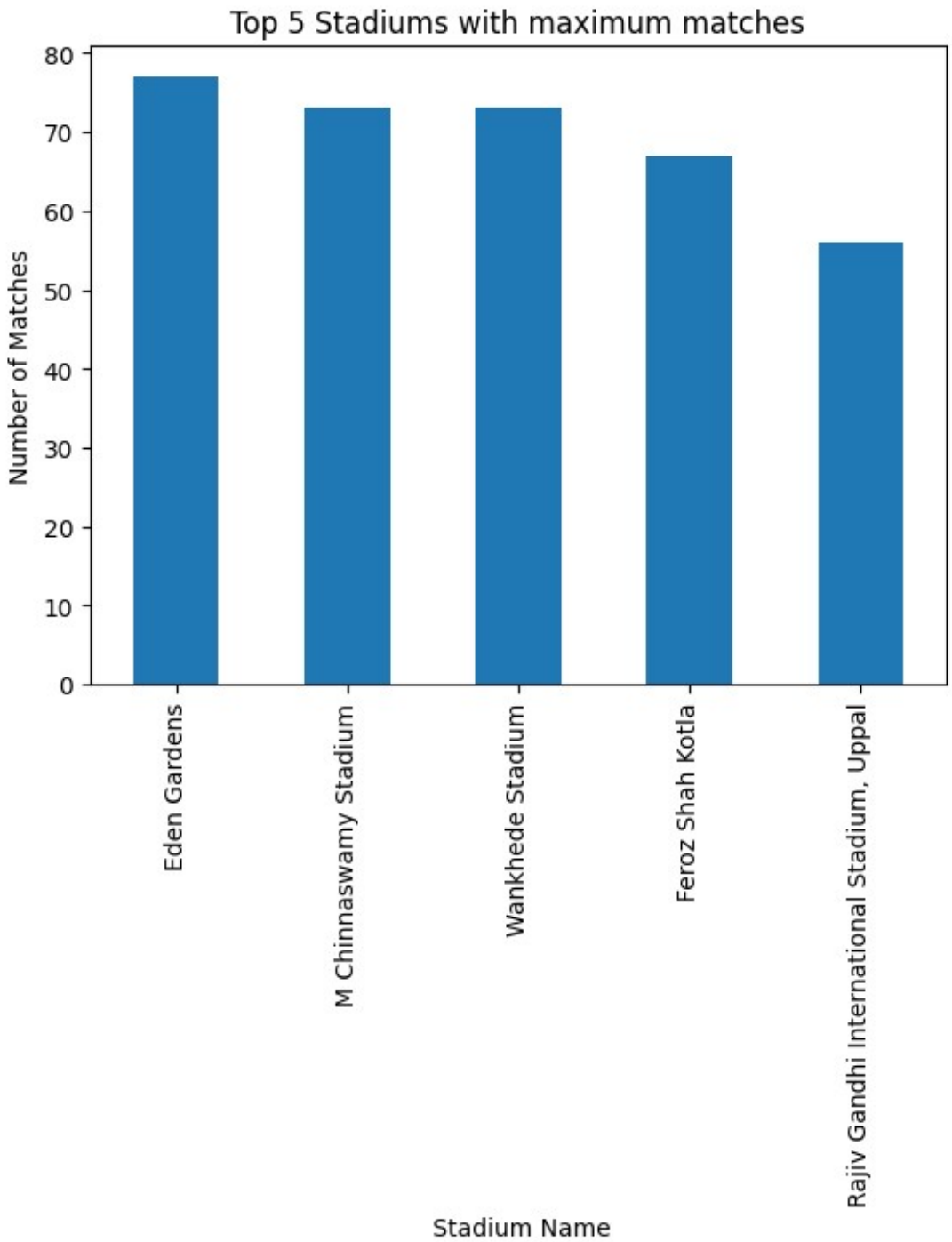
```
venue
Eden Gardens                                                77
M Chinnaswamy Stadium                                       73
Wankhede Stadium                                            73
Feroz Shah Kotla                                            67
Rajiv Gandhi International Stadium, Uppal                   56
```

```
MA Chidambaram Stadium, Chepauk                            49
Sawai Mansingh Stadium                                     47
Punjab Cricket Association Stadium, Mohali                 35
Maharashtra Cricket Association Stadium                    21
Dr DY Patil Sports Academy                                 17
Subrata Roy Sahara Stadium                                 17
Kingsmead                                                  15
Punjab Cricket Association IS Bindra Stadium, Mohali       14
SuperSport Park                                            12
Sardar Patel Stadium, Motera                               12
Brabourne Stadium                                          11
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium        11
Saurashtra Cricket Association Stadium                     10
Holkar Cricket Stadium                                      9
Himachal Pradesh Cricket Association Stadium                9
New Wanderers Stadium                                       8
M. A. Chidambaram Stadium                                   8
Rajiv Gandhi Intl. Cricket Stadium                          8
Newlands                                                    7
St George's Park                                            7
Barabati Stadium                                            7
JSCA International Stadium Complex                          7
Sheikh Zayed Stadium                                        7
Dubai International Cricket Stadium                         7
Feroz Shah Kotla Ground                                     7
M. Chinnaswamy Stadium                                      7
IS Bindra Stadium                                           7
Shaheed Veer Narayan Singh International Stadium            6
Sharjah Cricket Stadium                                     6
Nehru Stadium                                               5
Green Park                                                  4
Buffalo Park                                                3
De Beers Diamond Oval                                       3
Vidarbha Cricket Association Stadium, Jamtha                3
OUTsurance Oval                                             2
ACA-VDCA Stadium                                            2
Name: count, dtype: int64
```

Top 5 Statdiums with maximum matches

```
matches['venue'].value_counts().head().plot(kind='bar', title= 'Top 5
Stadiums with maximum matches')
plt.xlabel(' Stadium Name')
plt.ylabel('Number of Matches')
```

```
Text(0, 0.5, 'Number of Matches')
```


Top 5 Stadiums with maximum matches
Number of Matches
0
10
20
30
40
50
60
70
80
Eden Gardens
M Chinnaswamy Stadium
Wankhede Stadium
Feroz Shah Kotla
Rajiv Gandhi International Stadium, Uppal
Stadium Name